

Víctor Manuel Chávez Pérez

Professor–Researcher · Applied Physics, Data Science & Science Education

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National Researcher System · Candidate (SECIHTI) 2025–2029



PROFILE

I am a physicist from the University of Guadalajara and hold a PhD in Applied Physics from the University of Vigo. My research blends atmospheric and climate physics with large-scale data analysis: I study sudden stratospheric warmings (SSW), heat waves and their response under climate-change scenarios using models such as CMIP5 and WACCM.

Beyond basic research, I bring physics to real-world problems: I am developing a flood early-warning system for the Guadalajara Metropolitan Area that integrates weather radar, hydrological runoff modeling over high-resolution LiDAR terrain and remote sensing. My work also spans physical oceanography, remote sensing and applied optics.

I am also interested in how generative AI can transform science education: I design learning assistants and teaching materials that I evaluate with students themselves. Alongside this, I co-founded the Tlakakini Science & Technology Club, devoted to science outreach for children. I believe the best science is the kind that gets shared.

RESEARCH INTERESTS

Sudden stratospheric warmings · Stratospheric dynamics · General circulation models (CMIP5 / WACCM) · Climate change · Physical oceanography & remote sensing · Applied optics · Data science · High-performance computing · Science education & outreach · Generative AI in science education · Machine Learning applied to climate & atmospheric data · Student perception of AI-based learning tools

EDUCATION

- 2017 – 2024** **PhD in Applied Physics**
Universidade de Vigo · Spain
Thesis: Sensitivity of sudden stratospheric warmings to simulation and analysis conditions in models.
- 2011 – 2012** **MSc in Climate Science: Meteorology, Physical Oceanography & Climate Change**
Universidade de Vigo · Spain
Temporal and spatial variability of the Minho river plume (MODIS 250 m). Top honors.
- 2009 – 2011** **MSc in Hydrometeorology — Meteorology & Physical Oceanography**
Universidad de Guadalajara · Mexico
Seasonality and anomalies of oceanographic variables in the Gulf of Mexico via remote sensing.
- 2010 – 2011** **Diploma in Teaching Competencies**
Universidad de Guadalajara · Mexico
Pedagogical training for university teaching.
- 2000 – 2008** **BSc in Physics**
Universidad de Guadalajara · Mexico
Thesis: Ocean and coastal meteorology of Nayarit (San Blas, 1999–2002). Grade 95/100.

EXPERIENCE

- 2025 – present** **Lecturer "B"**
Centro Universitario de Tlaquepaque · UdeG · Tlaquepaque, Mexico
Department of Basic Sciences. Chair of the Basic Sciences Academic Board.
Courses: Probability · Electromagnetism Lab · Discrete Mathematics · Numerical Methods · Calculus · Advanced Calculus
- 2025 – 2026** **Lecturer**
Centro Universitario Bonpland y Humboldt · Online
PhD in Aerospace Engineering: Research Methodology and Fluid Mechanics.
Courses: Research Methodology · Fluid Mechanics
- 2019 – present** **Lecturer "B"**
Universidad Tecnológica de Jalisco · Guadalajara, Mexico
Mechatronics Division — Basic Sciences.
Courses: Engineering Physics · Engineering Mathematics I & II · Probability & Statistics · Integral Calculus · Applied Engineering Statistics · Thermodynamics

- 2019 – 2020 Faculty Lecturer**
Tecnilenio · Campus Guadalajara · Guadalajara, Mexico
High school: Matter & Energy, Integral Calculus, Science & Technology.
Courses: Matter & Energy · Integral Calculus · Science & Technology
- 2019 Physics Instructor**
UNITEC · Campus Guadalajara · Guadalajara, Mexico
Engineering programs: Statics.
Courses: Statics
- 2016 – 2024 External Collaborating Researcher**
Universidade de Vigo · EPhysLab · Ourense, Spain
Doctoral research on sudden stratospheric warmings in CMIP5 and WACCM models; statistical analysis of large climate datasets.
- 2015 Predoctoral mobility stay**
Universidad Pablo de Olavide · Seville, Spain
Short research stay at R&D centers; results presented at international conferences.
- 2013 – 2014 Predoctoral research stay**
Universidad de Guadalajara · Guadalajara, Mexico
Academic research stay on climate-change analysis.
- 2012 – 2016 Predoctoral Researcher (FPI Fellowship)**
Universidade de Vigo · EPhysLab · Ourense, Spain
Statistical analysis of large-volume data for climate-change research with general circulation models.
- 2012 – 2013 Physics & Math Instructor**
Universidad de Especialidades · Guadalajara, Mexico
Mechanics, Introduction to Physics and Electromagnetism.
Courses: Mechanics · Intro to Physics · Electromagnetism
- 2010 – 2011 Physics & Math Instructor**
Centro de Enseñanza Técnica Industrial (CETI Colomos) · Guadalajara, Mexico
Engineering programs.
Courses: Differential & Integral Calculus · Statics · Dynamics · Mathematics
- 2008 – 2011 Professor–Researcher**
Instituto Tecnológico Superior de Zapopan · Zapopan, Mexico
Mathematics for engineering; parallel computing applied to oceanography.
Courses: Physics (Mechanics, E&M) · Differential & Integral Calculus · Differential Equations · Dynamics

PUBLICATIONS

- 2025 Observación Astronómica en México: Pasando de lo Visible a lo Invisible**
Almaguer-Gómez C., Medina J. F., Chávez-Pérez V. M., et al. · *Óptica Pura y Aplicada* 58(1):15 · DOI: [10.7149/OPA.58.1.51200](https://doi.org/10.7149/OPA.58.1.51200)
- 2025 Influence of Satellite and Presatellite Periods on the Validation of Major Sudden Stratospheric Warmings in Historical CMIP5 Simulations**
Chávez-Pérez V. M., Añel J. A., et al. · *Atmosphere* 16(5) · DOI: [10.3390/atmos16050628](https://doi.org/10.3390/atmos16050628)
- 2025 Temporal Variability of Major Stratospheric Sudden Warmings in CMIP5 Climate Change Scenarios**
Chávez-Pérez V. M., et al. · *Climate* 13(10) · DOI: [10.3390/cli13100207](https://doi.org/10.3390/cli13100207)
- 2023 Estacionalidad y Anomalías del Nivel del Mar en el Caribe por Medio de Datos de Altimetría**
Palacios Hernández E., Chávez Pérez V. M., Sierra Romero L. L. · *Generis Publishing* · ISBN: 979-8-88676-642-4
- 2022 Impact of Increased Vertical Resolution in WACCM on the Climatology of Major Sudden Stratospheric Warmings**
Chávez-Pérez V. M., Añel J. A., et al. · *Atmosphere* 13(4) · DOI: [10.3390/atmos13040546](https://doi.org/10.3390/atmos13040546)
- 2018 The climatology of dust events over the European continent using data of the BSC-DREAM8b model**
Mandija F., et al. · *Atmospheric Research* 209:144–162 · DOI: [10.1016/j.atmosres.2018.03.006](https://doi.org/10.1016/j.atmosres.2018.03.006)
- In preparation**
- 2026 Science Beyond the Classroom: Impact Assessment of a Monthly Science Club During National Non-School Days in Mexico**
Almaguer-Gómez C., Chávez-Pérez V. M., et al.
- 2026 Student Perception of a Generative AI-Based Calculus Learning Assistant: A Mixed-Methods Study**
Chávez-Pérez V. M., et al.

2026 Assessment of the lightning activity response to aerosol optical depth
Mandija F., Chávez-Pérez V. M.

CONFERENCES & TALKS

- 2022** **LXV Congreso Nacional de Física**
2–7 oct 2022 · Zacatecas, Mexico
• Análisis de la frecuencia de ocurrencia de los calentamientos súbitos estratosféricos en el Hemisferio Sur en los modelos de altura estratosférica del CMIP5
• Caracterización óptica de resina transparente de impresión 3D
- 2020** **LXIII Congreso Nacional de Física**
4–9 oct 2020 · Online
• Impacto en las significancias de las características básicas de los SSW al comparar modelos del CMIP5 con datos de reanálisis en periodos pre-satelital y pos-satelital
• Impacto en la frecuencia de ocurrencia de los calentamientos súbitos estratosféricos principales en distintos escenarios de cambio climático usando CMIP5
- 2018** **American Geophysical Union (AGU) Fall Meeting 2018**
10–14 dic 2018 · Washington, USA
• Impact of Increased Vertical Resolution in WACCM on the Climatology of Major Stratospheric Sudden Warmings
- 2018** **Reunión Anual de la Unión Geofísica Mexicana**
28 oct–2 nov 2018 · Puerto Vallarta, Mexico
• Cambios en la climatología de los calentamientos súbitos estratosféricos principales usando WACCM con alta resolución vertical
• Evaluación de la capacidad de reproducción de los calentamientos súbitos estratosféricos principales en el conjunto de datos históricos de CMIP5
• Calentamientos súbitos estratosféricos principales bajo escenarios de cambio climático con CMIP5
- 2017** **European Geosciences Union (EGU) General Assembly 2017**
23–28 abr 2017 · Vienna, Austria
• Estimation of the variability of dust aerosol optical depth over several European regions based on forecast model data
- 2016** **LIX Congreso Nacional de Física (SMF)**
2–7 oct 2016 · León, Mexico
• Teleconexión estratosférica y los calentamientos súbitos estratosféricos
• Patrones característicos en las variables estratosféricas de la Oscilación Cuasi-Bienal
• Anomalías estratosféricas antes y después de los calentamientos súbitos estratosféricos para las distintas fases de la Oscilación Cuasi-Bienal
- 2016** **SPARC Workshop SHARP 2016 — Stratospheric Change and its Role for Climate Prediction**
11–15 jul 2016 · Berlin, Germany
• Changes in the stratosphere before and after SSW events in the different phases of the Equatorial QBO
• Differences in the detection and classification of the Stratospheric Sudden Warming over the three reanalyses for the period 1979–2014
- 2015** **European Geosciences Union (EGU) General Assembly**
12–17 abr 2015 · Vienna, Austria
• Variability of the Brewer–Dobson circulation during Stratospheric Sudden Warmings
- 2015** **SPARC Regional Workshop — Role of the stratosphere in climate variability and prediction**
12–13 ene 2015 · Granada, Spain
• Variability of the Brewer–Dobson circulation during Stratospheric Sudden Warmings
- 2010** **Reunión Anual de la Unión Geofísica Mexicana**
7–12 nov 2010 · Puerto Vallarta, Mexico
• Estacionalidad y anomalías de las variables de percepción remota oceanográficas en el Golfo de México
• Estacionalidad y anomalías de la superficie del nivel del mar en el Mar Caribe por medio de datos de altimetría
• Alternativa para el procesamiento de datos oceanográficos de gran cómputo mediante programación en paralelo con MatLab
- 2010** **LIII Congreso Nacional de Física**
25–29 oct 2010 · Boca del Río, Veracruz, Mexico
• Supercómputo aplicado a programación en paralelo para optimizar procesos en Astrofísica y Oceanografía Física
- 2009** **LII Congreso Nacional de Física (SMF)**
26–30 oct 2009 · Acapulco, Guerrero, Mexico
• Espectros de frecuencias de datos meteorológicos de Isla María Madre
• Comparación del viento costero y marino de la zona costera de San Blas, Nayarit (1999–2002)

ACTIVE RESEARCH PROJECTS

- 2026 – actual** **Flood early-warning system for the Guadalajara Metropolitan Area — Principal investigator**
Centro Universitario de Tlaquepaque de la Universidad de Guadalajara · Funding: No external funding
Development of an urban flood early-warning system that combines remote sensing (IAM-UdeG weather radar), hydrological runoff modeling over high-resolution LiDAR terrain, and official flood inventories. The system classifies risk by sub-basin in real time and estimates water arrival time, aiming to calibrate thresholds through a retrospective analysis with historical Civil Protection/Fire Department records.
- 2026 -** **Generative AI for lab practice reports (Google Gems) — Principal investigator**
Centro Universitario de Tlaquepaque de la Universidad de Guadalajara · Funding: No external funding
Implementation of Google Gems as a writing and feedback assistant for students to draft and self-assess their lab practice reports in physics courses.
- 2025 – actual** **AI-generated educational comics for science teaching — Principal investigator**
Universidad de Guadalajara · Funding: No external funding
Development and evaluation of educational comics created with generative AI tools as a didactic resource for teaching physics, mathematics and science at basic and upper-secondary levels.
- 2025 – actual** **Pedagogical Calculus Assistant with Google Gems: development and student perception — Principal investigator**
Universidad Tecnológica de Jalisco · Funding: Technological University of Jalisco
Design and implementation of a Calculus learning assistant based on Google Gems, with quantitative and qualitative study of students' perception, satisfaction and academic performance.
- 2025 – 2028** **Optical elements to enhance visual acuity via wavefront coding — Collaborator**
Universidad de Guadalajara · Funding: Universidad de Guadalajara
Design and validation of optical elements that improve visual acuity through wavefront coding.
- 2025 – actual** **World Wide Lightning Location Network (WWLLN) — Collaborator · antenna lead**
University of Washington · Funding: University of Washington
Global lightning location network. In charge of the receiving antenna/node and real-time data delivery.
- 2024 – actual** **Tlakakini Science & Technology Club — Coordinator**
CUCEI · Universidad de Guadalajara · Funding: University of Guadalajara · CUCEI
Science-outreach club for children, with monthly experiment and workshop sessions.

TECH DEVELOPMENTS

Flood early-warning system · Guadalajara Metro Area

Python · FastAPI · Leaflet · NumPy · pysheds · rasterio · GDAL · Docker

Real-time web platform that anticipates urban flooding in the Guadalajara Metropolitan Area. It fuses the IAM-UdeG weather radar (one scan per minute) with a hydrological runoff model built on 5 m LiDAR and Copernicus DEM terrain, plus the IMEPLAN official flood inventory (363 recurrent sites). It produces a per-zone risk light —calibrated against official water-depth records— and a continuous risk grid that routes rainfall downslope, with concentration time and a "water incoming" ETA. Includes live and historical-replay views. Deployed in production.

ACADEMIC SERVICE

Thesis supervision

- 2011** **Estacionalidad y anomalías del nivel del mar en el Caribe por medio de datos de altimetría**
Lilia Lizeth Sierra Romero · Bachelor's · Co-advisor · Universidad de Guadalajara

Journal peer review

- 2025** **IEEE Latin America Transactions**
Reviewer — Q2 journal, double-blind
- 2024–2025** **Int. J. of Environment and Pollution (Inderscience)**
Reviewer — Q4 journal, double-blind
- 2025** **Congreso de Ingeniería, Ciencia y Gestión Ambiental (AMICA)**
Scientific committee reviewer

Project evaluation

- 2025–2026** **SECIHTI**
External evaluator — 2026 National Call, Basic & Frontier Science (3 proposals)

SCIENCE OUTREACH

- 2024 –** **Tlakakini Science & Technology Club**
Co-founder. Monthly outreach sessions for children on national non-school days.

- 2009 – Physics conferences (AGU, EGU, SMF)**
Talks at national and international conferences on climate and the stratosphere.
- 2005 – Training science communicators**
Optics and electromagnetism workshops; National Outreach Meeting.
- 2009 – Environment & Sustainability Thematic Network**
Member of the CONACYT environment and sustainability network.
- 2011 – Environmental Physics Laboratory (EPHysLab)**
External collaborating researcher with the University of Vigo group.

GRANTS & HONORS

- 2025–2029 National Researcher System · Candidate**
SECIHTI · Mexico
- 2012–2016 FPI Grant · Predoctoral Researcher**
Ministerio de Educación · Spain
- 2011–2012 Climate Science Master's Scholarship**
Campus do Mar · Spain
- 2011–2012 «Ourense Exterior» mobility grant**
Diputación de Ourense · Spain
- 2009–2011 Hydrometeorology Master's Scholarship**
CONACYT · Mexico
- 2009 Admission to the State Research System**
COECYTJAL · Jalisco, Mexico

CONTINUING EDUCATION

- 2025 Diploma** — Estrategias tecnopedagógicas para el desarrollo de habilidades del siglo XXI · Universidad de Guadalajara
- 2025 Course** — Inteligencia Artificial para la docencia · Universidad de Guadalajara
- 2025 Course** — Inteligencia Artificial Generativa para Docentes · Universidad Anáhuac
- 2025 Course** — IA Generativa en el aula · UNAM
- 2025 Course** — Python for Data Science, AI & Development · IBM
- 2025 Certification** — What is Data Science? · IBM
- 2025 Course** — Cálculo Diferencial para Data Science e IA · Platzi
- 2025 Course** — Estadística y Probabilidad · Platzi
- 2025 Course** — Fundamentos de Ingeniería de Software · Platzi
- 2025 Course** — Introducción a la Terminal y Línea de Comandos · Platzi
- 2014 Course** — OOP in Atmospheric and Oceanic Model Development · Universidade de Vigo
- 2011 Diploma** — Competencias Docentes: Inducción al Bachillerato General · Universidad de Guadalajara
- 2005 Workshop** — Formación de Divulgadores de la Ciencia · Universidad de Guadalajara

TOOLBOX

Python · Fortran · MatLab · IDL · NCL · LaTeX · COMSOL · LabVIEW · MPI · Linux · Moodle · Pandas · Scikit-learn · Jupyter